ALICE L WALTON SCHOOL OF MEDICINE

Acute Lymphoblastic Leukemia/ Lymphoma

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Adapted from: Acute Lymphoblastic Leukemia/ Lymphoma

Learning Objectives (4) | Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of patients with acute lymphoblastic leukemia (ALL) from other benign and malignant disorders.
2. Explain the common genetic abnormalities and pathologic findings in ALL.
3. Interpret clinical findings to properly diagnose ALL.
4. Outline the

CASE CONNECTION

ML is a 10-year-old who comes to your office with her mother, GL. “She has been so tired for the past 3 months,” says GL. “She missed all her soccer practices, and she doesn’t even want to play her video games. We’ve also noticed that she bruises really easily.” ML’s vital signs are within normal limits, as is her physical examination. A complete blood count (CBC) shows a high white blood cell (WBC) count, low hemoglobin, and very low platelet count. A peripheral smear shows lymphoblasts.

How will you explain ML’s condition to her mother? Consider your answers as you read, and we’ll revisit ML at the end of the brick.

GO TO CONCLUSION ↓

What Is Acute Lymphoblastic Leukemia/Lymphoma (LO #1)?

Acute lymphoblastic leukemia/lymphoma (ALL) is a malignant neoplasm in which very early lymphocyte precursors called “lymphoblasts” proliferate in the bone marrow, lymph nodes, or blood.

In ALL, malignant lymphoblasts divide very quickly, filling up the bone marrow, crowding out normal hematopoietic precursor cells, leading to serious consequences for the patient if not treated immediately.

First, let’s see where this disorder with a somewhat confusing name fits into the larger picture of h

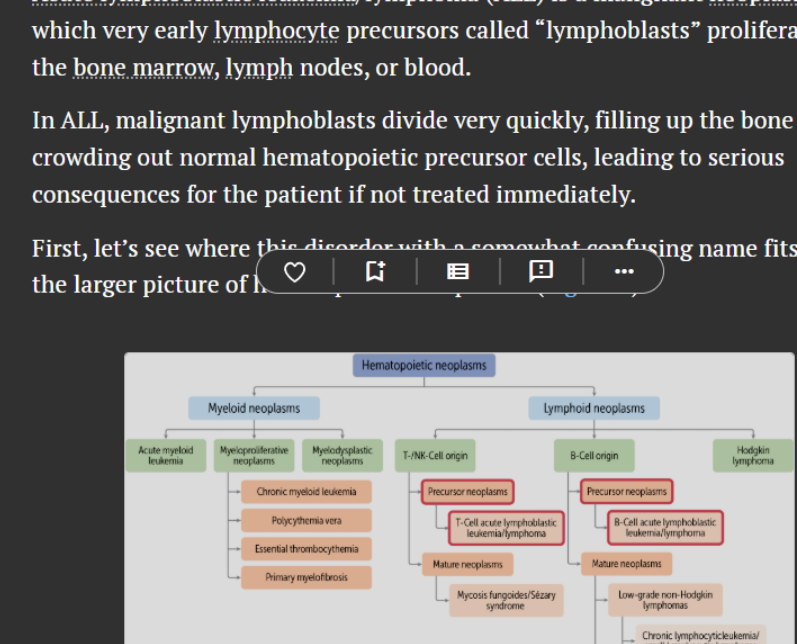


Figure 1

You’ll see that ALL is a lymphoid neoplasm (derived from lymphocyte rather than myeloid precursors), and that it can arise from either T- or B-cell precursor cells. The B-cell type is the most common (85% of cases) and has a better prognosis; T-cell ALL accounts for 15-20% of ALL cases. The two types are treated differently.

Leukemia or Lymph

Let’s discuss the confusion with the name. Aren’t leukemias and lymphomas different diseases? Yes, both are neoplastic hematopoietic diseases, but leukemias start in the bone marrow, and lymphomas start in lymph nodes.

So why do we lump leukemia and lymphoma together in this disease? After studying many cases of acute lymphoblastic leukemia and acute lymphoblastic lymphoma, it’s now known that the two diseases are really the same thing! How do we know? The cells:

- “Mark” the same (share similar cell surfaces and nuclear markers)
- Have the same prognosis
- Will eventually progress to involve all three sites (bone marrow, lymph nodes, and blood).

So, we no longer separate the two diseases but rather just lump them together as acute lymphoblastic leukemia/lymphoma. This is similar to what we do in low-grade B-cell non-Hodgkin lymphomas.

Why are acute lymphoblastic leukemia and lymphoblastic lymphoma regarded as the same disease?

Epidemiology

ALL is much more common in children than in adults and is the most common **childhood malignancy** overall (40 cases per million children), accounting for 25% of childhood cancers. In fact, if your patient is a child who presents with an acute leukemia, you assume it is ALL until proven otherwise. In children, ALL has a good prognosis.

ALL can occur in adults as well, but less commonly. When it does, the prognosis is poor.

What group is most affected by acute lymphoblastic leukemia/lymphoma?

How Do Patien ic Leukemia/Lymphoma Present (LO #1)?

Like any other acute leukemia or lymphoma, ALL typically presents over a short period of time (weeks) with serious symptoms.

Common Symptoms

Table 1 presents the most common signs and symptoms of ALL, as well as their underlying causes.

Table 1

Sign/Symptom	Cause
Bone/joint pain	Expanding blast population in the bone marrow
Fatigue/pallor	Anemia
Petechiae/easy bruising/bleeding	Thrombocytopenia
Fever	Infection, cytokines from leukemia

The signs of cytopenias (decreased blood cell counts) such as anemia and thrombocytopenia are due to replacement of bone marrow with neoplastic cells (no room to produce red blood cells or platelets).

There may be abdominal fullness due to an **enlarged spleen and/or liver**. These can be due to direct organ infiltration by neoplastic cells or because of extramedullary hematopoiesis (manufacture of blood cells in the liver and spleen when the bone marrow is replaced by leukemic cells).

Patients with a lymph opathy (lymph node enlargement). In particular, T-cell ALL commonly presents with lymph node or mediastinal involvement.

CLINICAL CORRELATION

T-cell ALL cases often present as an anterior mediastinal mass that can cause severe cardio-respiratory compromise due to the tumor pressing on cardiac and respiratory structures. In these cases, a biopsy may have to be done in the operating room to prepare for possible intubation.

Mnemonic

Anterior Mediastinal Mass Differential Diagnosis: the 4 T’s

What Is the Pathophysiology of Acute Lymphoblastic Leukemia/Lym

Most cases of ALL have no known cause, but there are some genetic links.

Etiology/Risk Factors

A few known associations are:

- Concurrent genetic syndromes such as Down syndrome, neurofibromatosis type 1, Fanconi anemia, or ataxia-telangiectasia. ALL is the most common leukemia associated with Down syndrome (trisomy 21).
- Family history of leukemia
- Ionizing radiation exposure

Genetics and Cell Biology

Acute lymphoblastic leukemia/lymphoma has associations with certain genetic mutations that contribute to their development. These mutations are important to identify as they affect treatment and prognosis. Table 2 outlines some important mutations (this is not a complete list).

Table 2.

Mutation	Occurrence	Prognosis
Hyperdiploidy (gain in chromosomes, >50)	25%	Good
t(12;21)	16-22%	Good
t(4;11)	60% infant ALL; ~5% other ALL	Poor
t(9;22)	20% adult ALL; 3-5% child ALL	Intermediate

CLINICAL CORRELATION

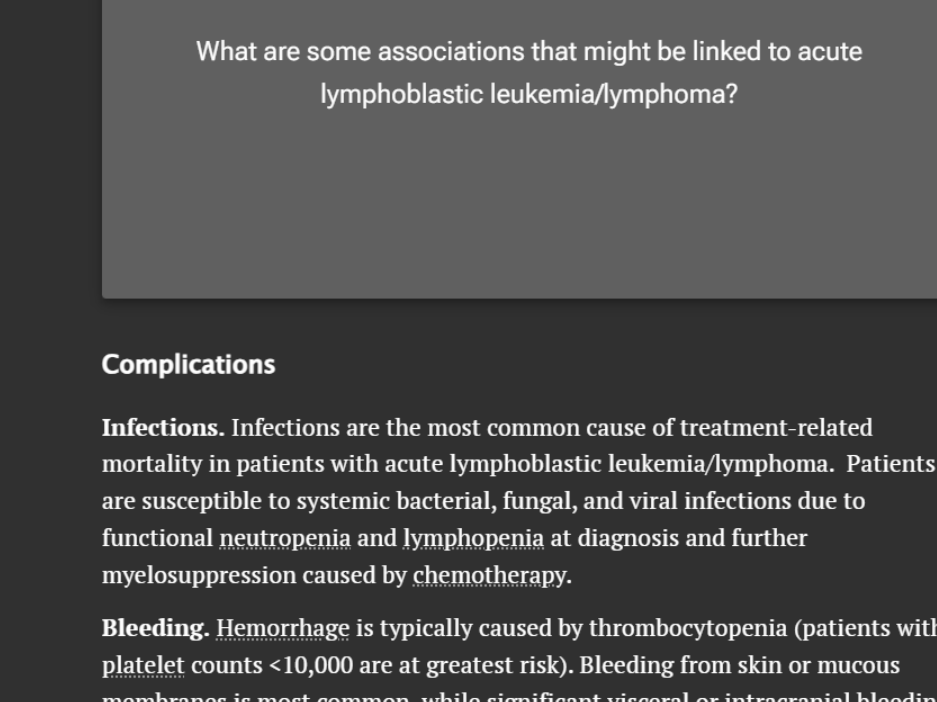
In general, prognosis in adult ALL is worse than children with ALL. One of the reasons is that children are more likely to have the favorable genetic mutations, and adults are more likely to have unfavorable mutations (such as t(9;22)).

CLINICAL CORRELATION

t(9;22), a translocation that leads to the development of the Philadelphia chromosome, is also seen in chronic myeloid leukemia (CML). The Philadelphia chromosome in CML, however, occurs at a different breakpoint region on chromosome 22.

Pathology

In ALL, both the lymph nodes and bone marrow can become filled with neoplastic **lymphoblasts.** These demonstrate large nucleoli, fine chromatin, and sometimes a small amount of bluish cytoplasm that may appear somewhat granular (Figure 2).



QUIZ

Figure 2

Blasts can also appear in the **peripheral circulation**, as well as in masses in the mediastinum, skin, or bone. Note that unlike the myeloblasts seen in acute myeloid leukemia and myelodysplastic syndrome, Auer rods (crystalline cytoplasmic inclusions) are not present in ALL.

What are some associations that might be linked to acute lymphoblastic leukemia/lymphoma?

Complications

Infections. Infections are the most common cause of treatment-related mortality in patients with acute lymphoblastic leukemia/lymphoma. Patients are susceptible to systemic bacterial, fungal, and viral infections due to functional neutropenia and lymphopenia at diagnosis and further myelosuppression caused by chemotherapy.

Bleeding. Hemorrhage is typically caused by thrombocytopenia (patients with platelet counts <10,000 are at greatest risk). Bleeding from skin or mucous membranes is most common, while significant visceral or intracranial bleeding is rare.

CNS Involvement. Blasts are most commonly found in the bone marrow, peripheral blood, and lymph nodes. However, there can be deposition in other body organs, causing severe symptoms. For example, ALL has a high propensity to disseminate to the ch is very concerning since the CNS has built-in barriers that make it resistant to some forms of chemotherapy. A similar problem occurs with testicular deposition of lymphoblasts.

CLINICAL CORRELATION

Symptoms of CNS involvement by leukemia include headache, vomiting, lethargy, and/or neck rigidity.

As the risk of CNS involvement by ALL is so high (especially for certain patients), institutions may give prophylactic intrathecal (delivered into the subarachnoid space) chemotherapy.

Tumor Lysis Syndrome. Tumor lysis syndrome (TLS) is an oncologic emergency that is caused by lysis of tumor cells that releases large amounts of potassium, phosphate, and nucleic acids into the circulation. Catabolism of the nucleic acids to uric acid leads to hyperuricemia.

How does this occur?

This process typically occurs after the initiation of cytotoxic chemotherapy but can occur spontaneously and with tumors that have a high proliferative rate, large tumor burden, or high sensitivity to cytotoxic therapy.

CLINICAL CORRELATION

Symptoms of TLS la rmalities and include nausea, vomiting, diarrhea, lethargy, hematuria, heart failure, cardiac arrythmias, seizures, muscle cramps, syncope, and possible sudden death.

One of the main concerns is precipitation of uric acid in the renal tubules, causing acute renal failure.

IV hydration and allopurinol (allopurinol blocks production of uric acid) are given to prevent tumor lysis syndrome. If the uric acid level is already elevated, rasburicase is administered. Rasburicase is a recombinant urate-oxidase enzyme which converts uric acid to allantoin (an inactive and soluble metabolite of uric acid).

How Do We Diagnose Acute Lymphoblastic Leukemia/Lymphoma (LO #3)?

Patients presenting with symptoms and signs of ALL (eg, fever, pancytopenia, lymphadenopathy, hepatosplenomegaly) should first receive a complete blood count (CBC), followed by either bone marrow biopsy or lymph node biopsy.

Complete Blood Count

In ALL the total WBC count may be decreased, normal, or increased. In some children the WBC count may increase to 50,000/mm^3 or more. Lymphoblasts may be seen on the blood smear.

CLINICAL CORRELATION

A diagnosis of leukemia generally requires **20% or more blasts in the blood or bone marrow**. There are some exceptions which will be mentioned in later bricks.

There may be decreased RBC and platelets, due to replacement of marrow with malignant blasts.

Biopsy

If lymphadenopathy is present, a lymph node biopsy can detect the abnormal lymphoblasts. Bone marrow biopsy and aspirate are essential if blood counts are abnormal on the CBC (Figure 3).

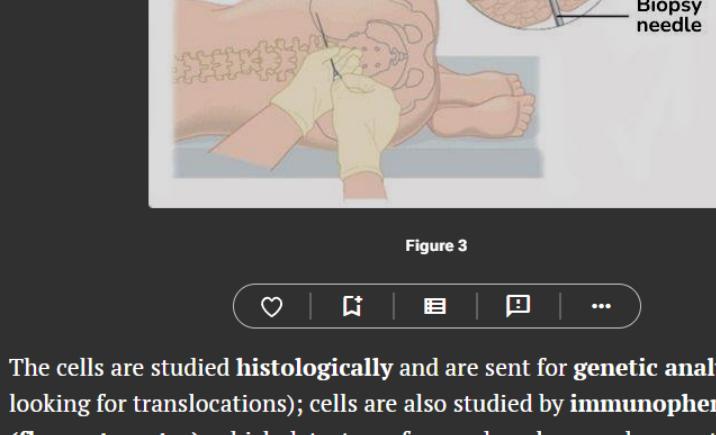


Figure 3

The cells are studied **histologically** and are sent for **genetic analysis** (eg, looking for translocations); cells are also studied by **immunophenotyping (flow cytometry)**, which detects surface and nuclear markers on the malignant cells.

There are a number of specific CD markers detected on the surface of B and T cells in ALL. These can be tested on peripheral blood or tissue samples and will help in selection of treatment and determining prognosis. Among the most common CD findings are:

- B-cell ALL: usually positive for CD19, CD22, and CD79a
- T-cell ALL: usually positive for CD2, CD3, CD4, CD7, and CD8

Detection of terminal deoxynucleotidyl transferase (TdT) markers, as expressed by lymphoblasts, is also useful to confirm the diagnosis of ALL. TdT markers are present in the nucleus of lymphoblasts and disappear as the cell matures into a normal lymphocyte.

Cytogenetic testing is used to guide management and prognosis. This can detect genetic abnormalities such as the 9;22 translocation (Philadelphia chromosome) discussed above.

CLINICAL CORRELATION

The above studies mentioned can also be performed on peripheral blood.

Laboratory Studies

Laboratory studies are usually done to evaluate for potential complications of ALL, such as coagulation abnormalities, liver or kidney dysfunction, metabolic abnormalities, and tumor lysis syndrome.

As such, here are some commonly ordered laboratory tests:

- Coagulation profile: Prothrombin time (PT), partial thromboplastin time (PTT), fibrinogen
- Chemistries: Electrolytes, renal and liver function tests, calcium, phosphate, LDH, uric acid

Lumbar Puncture/Neurologic Evaluation

Patients with ALL undergo lumbar puncture (LP) prior to initiating therapy to evaluate potential CNS involvement and concomitantly administer the first dose of intrathecal therapy (Figure 4).

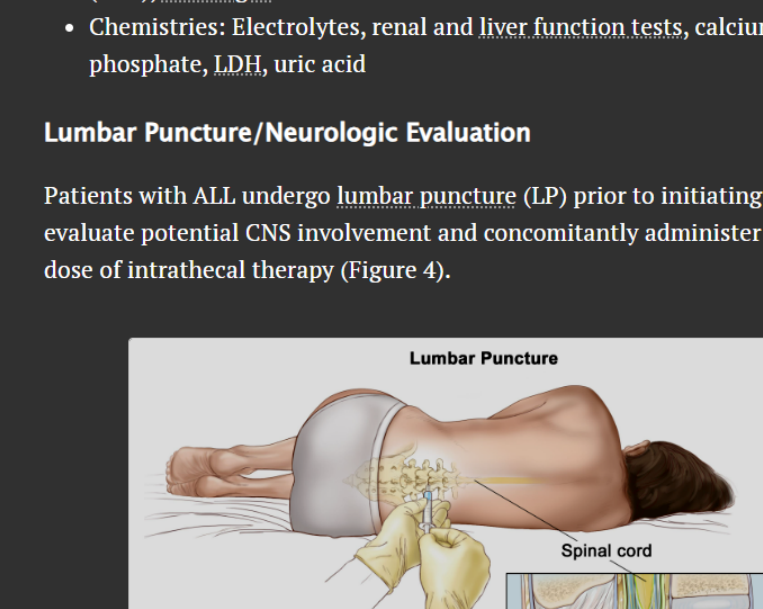


Figure 4. Lumbar Puncture.

Neurologic findings from the history and physical exam should be evaluated promptly by imaging (MRI or CT).

Imaging Studies

Imaging studies are helpful in looking for distant spread of the disease. Figure 5 (left) shows a mediastinal lymphadenopathy from the chest x-ray of a 2-year-old girl with T-cell ALL, which can be compared to a normal x-ray in Figure 5 (right).

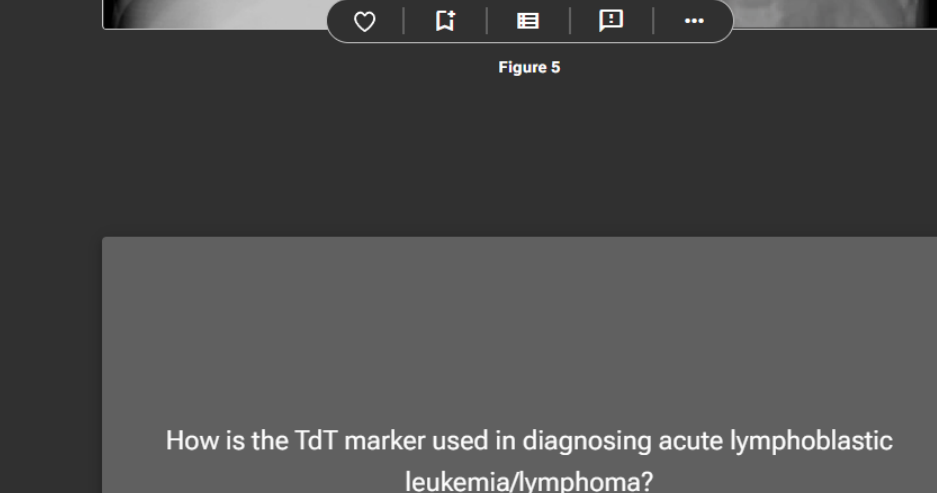


Figure 5

How is the TdT marker used in diagnosing acute lymphoblastic leukemia/lymphoma?

How Do We Treat Acute Lymphoblastic Leukemia/Lymphoma (LO #4)?

Treatment includes systemic chemotherapy as well as radiotherapy if there is disease in the CNS or testicles. The majority of the therapy is done on an outpatient basis and lasts 2.5 years.

Systemic Therapy

The drugs used to tre apy drugs as part of **multiple-drug chemotherapy.** The first goal of therapy is to induce complete remission and to restore normal hematopoiesis. The **induction therapy** regimen used includes a combination of agents. A typical regimen for children includes a glucocorticoid (prednisone, prednisolone, or dexamethasone), vincristine, daunorubicin, and L-asparaginase. Induction therapy duration on average is between *4-8 weeks.*

Glucocorticoids: Dexamethasone has a longer half-life and penetrates the CNS better than prednisone. However, use of dexamethasone can cause long-term *steroid-associated ADRs* such as fractures, osteonecrosis, infections, and mood/behavior problems.

Vincristine: *Mechanism:* Inhibition of microtubule function. Dose-related *peripheral neuropathy* is associated with use, which can cause paresthesia, loss of reflexes, weakness, and autonomic paralysis (including vocal cord paralysis).

Asparaginase: *Mechanism:* Asparaginase *Erwinia chrysanthemi* is an asparagine-specific enzyme derived from *Erwinia chrysanthemi* (Gram-negative *bacteria).* This enzyme catalyzes the conversion of the amino acid L-asparagine into aspartic acid and ammonia, reducing circulating asparagine levels. Leukemic cells have an inability to synthesize asparagine due to a lack of asparagine synthetase activity. Asparaginase deprives leukemic cells of an exogenous source of asparagine, ultimately leading to cell death.

What are the adverse effects? Asparaginase can cause infusion reactions (>10%), allergic reactions, coagulopathies, acute pancreatitis (4%), and hepatic toxicity (>10%).

What is the mechanism of action of Asparaginase?

Daunorubicin: *Mechanism:* Inhibition of DNA/RNA synthesis. Cumulative dose-related cardiomyopathy can occur, which can also cause heart failure.

Philadelphia chromosome (Ph)-positive ALL are in <5% of children, however the incidence is higher in adolescent and adult patients. In cases of Ph-positive ALL, a TKI (Tyrosine Kinase Inhibitor) is included. TKIs work by competitively inhibiting the BCR-ABL tyrosine kinase function that results from the BCR-ABL gene. *Which TKIs are included in therapy?* Long term treatment with Dasatinib-based therapy have shown better outcomes than Imatinib.

Consolidation therapy: Consolidation therapy for ALL lasts about *4-8 months*. Improvements in effective immunotherapy, along with evolving regimens, have led to enhanced outcomes. A typical **consolidation regimen** includes glucocorticoids, cyclophosphamide (alkylating agent), cytarabine, vincristine, asparaginase, and 6-mercaptopurine (purine antagonist).

Maintenance therap for at least *2 years* and includes the use of methotrexate, vincristine, and glucocorticoids. Doses, schedules, and duration of therapy can vary based on the patient.

Both children and adult patients with ALL are at increased risk for a second malignancy, particularly brain tumors and acute myelogenous leukemia, secondary to the chemotherapy treatment.

In case of **CNS infiltration, triple therapy** is used. These agents are directly administered to the subarachnoid space with a lumbar puncture. The agents used are

- **Methotrexate**-Mechanism: Competitive inhibition of dihydrofolate reductase, inhibiting DNA synthesis, repair, and replication. Adverse effects include myelosuppression, nephrotoxicity, hepatotoxicity, and mucositis.
- **Cytarabine**: Mechanism: A pyrimidine analog that is incorporated into DNA synthesis and inhibits DNA polymerase function. Adverse effects include myelosuppression and megaloblastic anemia
- **Hydrocortisone** (glucocorticoid)

Prognosis

When ALL occurs in children, the overall prognosis is good. With current therapy, most children achieve a complete and durable remission that lasts at least 5 years.

The prognosis in adults is less favorable. Only 20% to 40% are cured, and the 5-year survival rate is only 25% to 50%.

B-lineage ALL tends t age ALL.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to ML, how do you explain her condition to her mother?

You tell ML’s mother that she has ALL, a leukemia seen in young children and the most common childhood malignancy. ML’s mother is understandably concerned. You tell her that with the right chemotherapy treatment, the prognosis is good. In fact, most children achieve a remission that lasts at least 5 years. ML and her mother consult a hematologist-oncologist for more information and to schedule treatment.

Summary

- Acute lymphoblastic leukemia/lymphoma (ALL) is a disease in which malignant lymphoblasts quickly proliferate in the bone marrow, blood, and lymph nodes.
- It is the most common childhood cancer.
- ALL is classified as B-cell ALL and T-cell ALL, depending on the progenitor cell of origin.
- ALL may present with lymphadenopathy, fever, fatigue, and signs and symptoms of bone marrow failure, including dyspnea and fatigue (from anemia), infections (from leukopenia), and bleeding (from thrombocytopenia).
- ALL is associated with Down syndrome and multiple translocations, including the t(9;22) translocation (Philadelphia chromosome) in adults.
- The bone marrow replaced by sheets of lymphoblasts, and there are often decreased numbers of the other cell lines, leading to anemia and bleeding.
- The disease can spread to the liver, spleen, bones, brain, and testicles.
- Diagnosis is confirmed by bone marrow or lymph node biopsy.
- The blasts may show characteristic CD markers or a marker called TdT (terminal deoxynucleotidyl transferase) that can help with treatment and prognosis.
- Complications of ALL include infection, bleeding, CNS involvement, and tumor lysis syndrome.
- Patients are treated with systemic therapy (including chemotherapy) and sometimes with radiation to the CNS or testicles.
- Children with ALL tend to have a good prognosis, but the 3-year survival rate for adults is only 25% to 50%.
- B-lineage ALL tends to have a better prognosis than T-lineage ALL.
- The treatment of ALL involves several phases of chemotherapy: induction (with a typical regimen including glucocorticoid, vincristine, daunorubicin, and L-asparaginase), consolidation, and maintenance chemotherapy. Medications may also be given into the CNS to either prevent or treat CNS infiltration.

Review Questions

1. Which of the following genetic mutations is common in adult patients with acute lymphoblastic leukemia/lymphoma, but not in pediatric cases?

A. ○ Hyperdiploidy
B. ○ Hypodipl
C. ○ t(4;11)
D. ● t(9;22)
E. ○ t(12;21)

Hide Explanation

The correct answer is t(9;22) (D), also known as the Philadelphia chromosome. Hyperdiploidy (A), t(4;11) (C), and t(12;21) (E) are frequently associated with pediatric ALL but not with adult cases. Hypodiploidy (B) is not a common genetic mutation in acute lymphoblastic leukemia/lymphoma.

2. Which of the following is least characteristic of the lymphoblasts seen in acute lymphoblastic leukemia/lymphoma?

A. ● Auer rods
B. ○ Fine chromatin
C. ○ Large nuclei
D. ○ Little cytoplasm
E. ○ Visible nucleoli

Hide Explanation

The correct answer is Auer rods (A), which are a marker of myeloid precursors and particularly suggestive for acute myelocytic leukemia or myelodysplastic syndromes, but not acute lymphoblastic leukemia/lymphoma (ALL). The blasts characteristic of ALL show fine chromatin (B), large nuclei (C), and little cytoplasm (D), leading to a high nuclear/cytoplasmic ratio; and visible nucleo (E).

3. A 50-year-old male comes to the clinic with increasing fatigue and easy bruising. The physical exam shows hepatosplenomegaly and multiple bruises. The complete blood count shows a normal white blood cell count, reduced hemoglobin, and the presence of lymphoblasts on the peripheral blood smear. Analysis of the lymphoblasts shows the presence of a 9;22 translocation. The presence of this translocation indicates that the patient may respond to which of the following medications?

A. ○ Asparaginase
B. ○ Corticosteroids
C. ● Dasatinib
D. ○ Radiotherapy
E. ○ Vincristine

Hide Explanation

The correct answer is dasatinib (C), which is a tyrosine-kinase inhibitor. This class of drugs is used in malignancies (like some adult acute lymphoblastic leukemia/lymphoma (ALL) where there is a 9;22 translocation (Philadelphia chromosome). The other medications listed are general chemotherapy drugs not related to this translocation, including asparaginase (A), corticosteroids (B), and vincristine. (E) Radiotherapy (D) may be used for central nervous system (CNS) or testicular treatment in any type of ALL.

4. What is a common clinical presentation specific to T-cell ALL?

A. ○ Abdominal fullness
B. ○ Bone pain
C. ○ Hepatomegaly
D. ● Mediastinal involvement
E. ○ Splenomegaly

Hide Explanation

The correct answer is mediastinal involvement (D). T-cell ALL commonly presents with lymph node or mediastinal involvement. Abdominal fullness (A) can be due to an enlarged spleen or ne pain (B) can occur in ALL but is no (C) and splenomegaly (E) are symptoms of organ infiltration or extramedullary hematopoiesis, not specific to T-cell ALL.

5. What diagnostic method is used to deliver intrathecal chemotherapy?

A. ○ Bone marrow biopsy
B. ● Lumbar puncture
C. ○ Lymph node biopsy
D. ○ MRI of the brain
E. ○ Venipuncture

Hide Explanation

The correct answer is lumbar puncture (B), in which a needle is placed into the subarachnoid space (location of CSF) to sample CSF fluid and deliver intrathecal chemotherapy. Bone marrow biopsy (A) is used to assess the status of the bone marrow. Lymph node biopsy (C) is used to evaluate pathology of a given lymph node. MRI of the brain (D) would evaluate the brain and spine for pathologies that can be seen radiologically but is not used in itself to deliver intrathecal chemotherapy. Venipuncture (E) would get you a peripheral blood sample, not the location for delivering intrathecal chemotherapy.

How would you rate this Brick?

References

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- Overview of the clinical presentation and diagnosis of acute lymphoblastic leukemia/lymphoma in children - UpToDate
- Treatment of acute lymphoblastic leukemia/lymphoma in children and adolescents - UpToDate
- Update on Lymphoblastic Leukemia/Lymphoma - ScienceDirect

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

Hematopathology. 4th ed. CRC Press; 2021.
- Liu D. *Tumors and Cancers: Endocrine Glands – Blood – Marrow – Lymph.* 1st ed. CRC Press; 2017.
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Review Learning Objectives

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01 ☑ ~~Differentiate the clinical presentation of patients with acute lymphoblastic leukemia (ALL) from other benign and malignant disorders.~~
02 ☑ ~~Explain the common genetic abnormalities and pathologic findings in ALL.~~
03 ☑ ~~Interpret clinical findings to properly diagnose ALL.~~
04 ☑ ~~Outline the management plan of ALL.~~

Objective Confidence: 4/4